DATA WAREHOUSING & DATA MINING (CSI3010) - LAB PROJECT REVIEW 1

AgroVision-Mine

A Multimodal Data Warehouse and Visual Pattern Mining Framework for Crop Disease and Pest Intelligence

VERIFIED REVIEW 1 STATE: 36 Authentic Field Images (Rice) | 4 Diagnostic Tests | 34 Visual Descriptors | Kimball Star Schema | 5 Real SQL OLAP Ops | 56/56 Tests Passed (100%)

PROJECT TEAM DETAILS

Team Members: [Student Name 1 - Reg. No.] | [Student Name 2 - Reg. No.] | [Student Name 3 - Reg. No.]

Faculty Guide: [Faculty Name] | Lab Slot: [Lab Slot] | Academic Term: 2026

Research Motivation: The Agricultural Data Gap

LIMITATIONS OF CURRENT APPROACHES

• Isolated Point Classifiers: Standard deep learning (CNNs/YOLO) treats foliar disease identification as an isolated inference task without data warehousing.
• Absence of Relational Rigor: Images reside in flat, disconnected directories. No referential integrity linking images to botanical taxonomy, cryptographic digests, or extracted features.
• Inability to Perform OLAP Exploration: Researchers cannot run aggregate queries across agronomic hierarchies (e.g. comparing lesion texture of bacterial vs fungal pathogens).
• Lack of Explainable Lineage: Black-box neural embeddings cannot explain why an image exhibits specific texture or foliar discoloration patterns.

THE AGROVISION-MINE SOLUTION

✓ Multimodal Data Warehouse: First-of-its-kind architecture applying Ralph Kimball's Star Schema methodology to agricultural imagery and classical visual features.
✓ Explainable Feature Store: 34 classical descriptors (RGB/HSV color moments, Haralick GLCM textures, morphometric shape contours) in a 44-column analytical dataset.
✓ SQL-Backed OLAP Analytics: Executes real SQL queries across dimensions: Roll-up to crop categories, True Drill-down to diseases, Slicing by crops, Dicing, and Pivoting.
✓ Enterprise Data Integrity: Surrogate keys, native SQLite foreign key enforcement (PRAGMA foreign_keys = 1), and idempotent ETL guarantee 100% database consistency.

Core Research Question: How can image-based crop disease observations be transformed into a multimodal analytical warehouse for explainable multidimensional decision support?

Project Objectives & Multimodal Pipeline Flow

1. Ingestion & Anti-Fabrication
Ingest authentic ICAR Indian agricultural imagery; derive botanical metadata dynamically without synthetic fabrication.

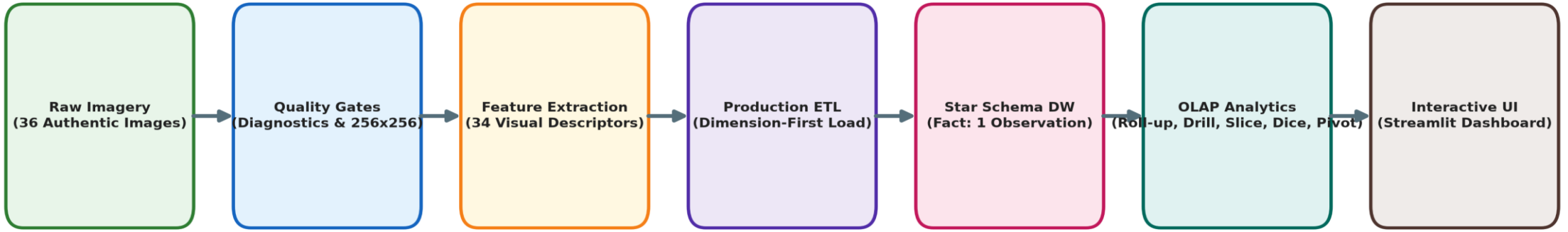
2. Quality Gate Auditing
Enforce automated validation: detect corrupted headers, exact MD5 duplicates, and sub-threshold images without silent deletion.

3. Classical Feature Store
Extract 34 explainable visual descriptors (Color moments, Haralick GLCM, morphometric contours) with full Image_ID lineage.

4. Kimball Star Schema
Deploy ANSI SQL relational schema centered on individual image observations surrounded by 4 empirically-supported dimensions.

5. Production Idempotent ETL
Build dimension-first ETL pipeline enforcing native SQLite foreign keys (PRAGMA foreign_keys = 1) with 0 duplicate fact re-runs.

6. Real SQL-Backed OLAP
Implement and evaluate genuine SQL analytical queries: Roll-up, True Drill-down, Slice, Dice, and Pivot over physical facts.



Dataset Description & Agronomic Taxonomy

Dataset Property	Verified Project Specification
Dataset Name	Official Govt. of India / ICAR Working Pathology Subset
Primary Source	data.gov.in / ICAR Research Institutes (ICAR-CRIDA, ICAR-IIRR)
Verified Working Subset	36 authentic field photographs across 3 foliar rice pathologies
Pathology Breakdown	Bacterial Leaf Blight: 12 | Brown Spot: 12 | Leaf Smut: 12
Host Plant Taxonomy	Rice (Oryza sativa) | Category: Cereal (National Staple Food)
Format & Standardization	Standardized 256x256x3 RGB (Lanczos anti-aliased interpolation)
Storage & Integrity	Unique Image_ID, MD5 cryptographic hash, Laplacian blur score

AGRONOMIC SIGNIFICANCE & DATA GOVERNANCE

- Critical Staple Crop: Rice feeds >60% of India's population. Microbial infections (Xanthomonas oryzae, Bipolaris oryzae) cause devastating seasonal yield losses of up to 40%.
- Authentic Working Subset: The 36 authentic observations represent the verified project subset from official Indian repositories, ensuring full reproducible auditability.
- Control Baseline Policy (HEALTHY): HEALTHY is maintained in dim_disease as an unpopulated control reference value (0 facts). The working subset contains pathological foliar specimens exclusively; no synthetic healthy observations were fabricated.
- Anti-Fabrication Mandate: Unsupported dimensions (Pest, Time, Geography) were strictly excluded because source images lack verified empirical field tags.

Image Preprocessing & Quality Gate Verification

CONTROLLED DIAGNOSTIC EDGE HARNESS (data/diagnostic_tests/)

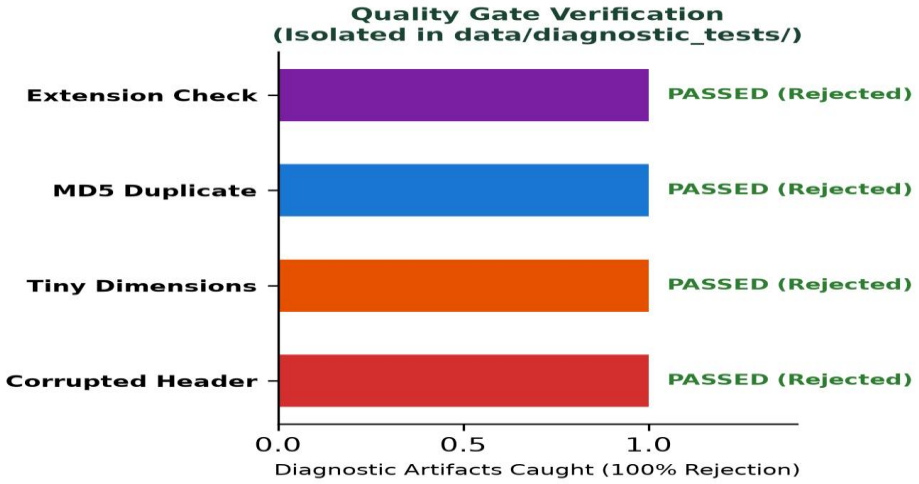
- corrupted_header_test.jpg -> Truncated byte stream -> REJECTED (CORRUPTED)
- invalid_dim_tiny.jpg -> Dimensions 16x16px < 32px -> REJECTED (INVALID_DIM)
- duplicate_exact_test.jpg -> Exact MD5 match -> REJECTED (EXACT_HASH)
- unsupported_extension.txt -> Non-image MIME -> REJECTED (UNSUPPORTED_FORMAT)

AUTHENTIC RESEARCH DATASET STANDARDIZATION

✓ 36 Authentic Field Images Discovered: 36 Successfully Standardized
✓ Quality Pass Rate: 100% (Zero authentic research observations rejected)
✓ Standardization: 256 x 256 px RGB, Lanczos anti-aliased resampling
✓ Non-Destructive Storage: Raw originals preserved; cleaned copies in data/processed/
✓ Laplacian Variance Blur Scoring: Mean score = 185.34 (Threshold = 100.0)







Visual Feature Extraction & Attribute Summary

Attribute Name	Physical Type	Agronomic / Visual Description	Role in DW
observation_key	INTEGER	Surrogate primary key for fact record	Fact PK
crop_key, disease_key	INTEGER	Surrogate foreign keys to botanical & pathology dims	Fact FKs
rgb_r/g/b_mean, std	FLOAT [0-255]	First and second order color moments across RGB channels	Feature Dim
hsv_h/s/v_mean, std	FLOAT	Hue, Saturation, Value moments for illumination invariance	Feature Dim
exg_index_mean	FLOAT	Excess Green Vegetative Index: 2G - R - B (chlorophyll intensity)	Feature Dim
dominant_color_tone	VARCHAR	Macroscopic foliar tone (Brown_Blight, Yellowish_Lesion)	Feature Dim
glcm_contrast, homogeneity	FLOAT	Haralick GLCM texture disruption & local smoothness	Feature Dim
lesion_severity_score	FLOAT [0-10]	Continuous composite foliar lesion severity measure	Fact Measure

44-COLUMN FEATURE DATASET

34 Explainable Visual Descriptors

- 17 Color Descriptors (RGB/HSV moments & ExG index)
- 7 Haralick GLCM Textures (Contrast, Energy, ASM, Homogeneity, Correlation, Dissimilarity)
- 7 Morphometric Shape Contours (Area, Perimeter, Aspect Ratio, Solidity, Coverage)

10 Lineage / Metadata Fields

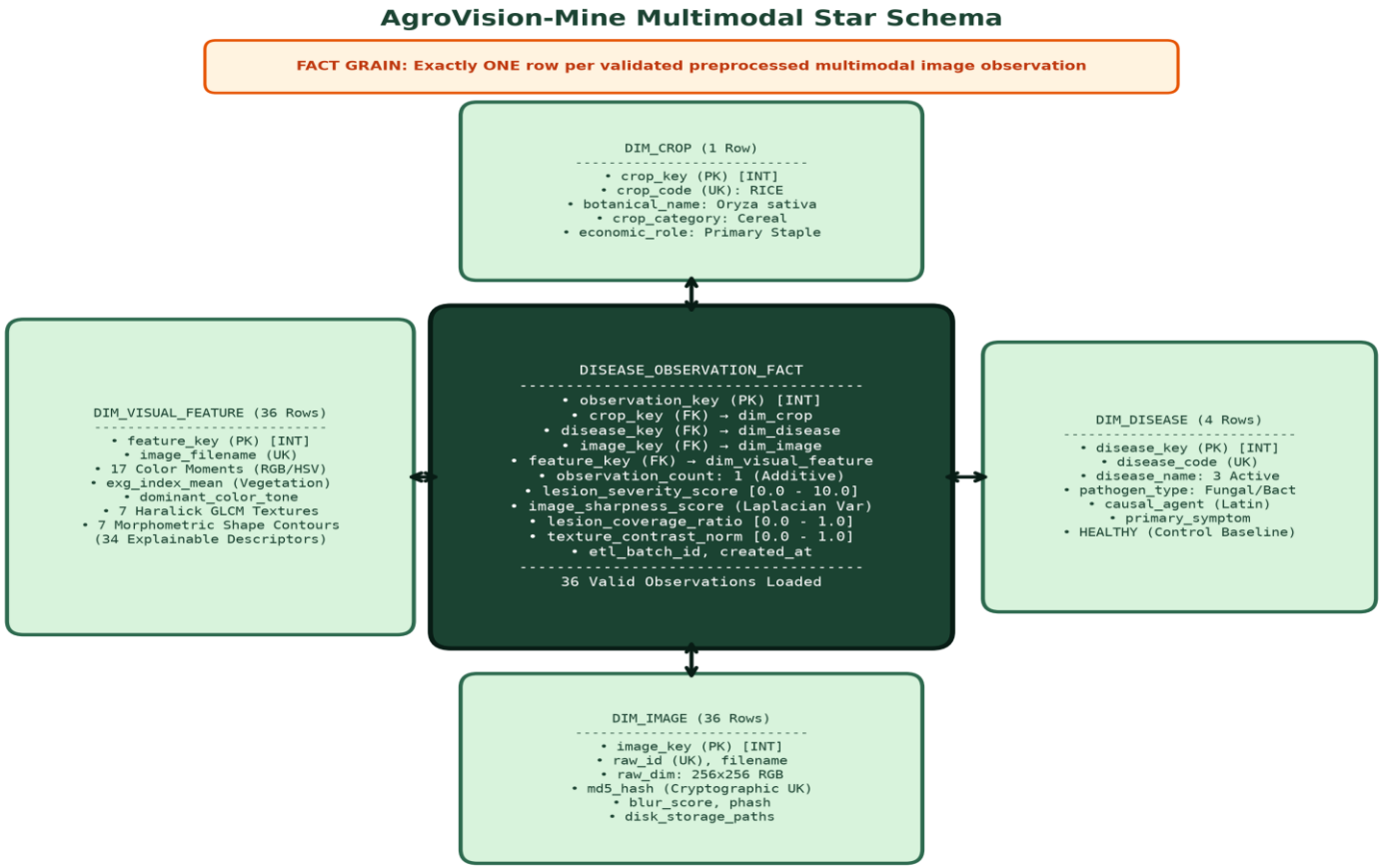
- image_id, filename, raw_path, processed_path, crop_label, condition_label, disease_label, pest_label, is_pest, md5_hash

100% Traceability

- Every row maps 1:1 to an authenticated image via image_id and MD5 cryptographic hash.

Standard Terminology Verification: 34 explainable visual descriptors stored in a 44-column analytical feature dataset (data/features/image_features.csv).

Multimodal Data Warehouse - Star Schema Topology



PHYSICAL WAREHOUSE INVENTORY

- **disease_observation_fact (36 Rows)**

PK: observation_key

- FKs: crop_key, disease_key, image_key, feature_key
- Measures: count (1), severity [2.13-4.98], sharpness [5.9-789.8], coverage [0.21-0.50], GLCM contrast

- **dim_crop (1 Row)**

Host taxonomy: RICE (Oryza sativa), Cereal

- **dim_disease (4 Rows)**

Bacterial Leaf Blight, Brown Spot, Leaf Smut

- HEALTHY (Unpopulated Control Baseline)

- **dim_image (36 Rows)**

Optical geometry, MD5 hash, blur score, paths

- **dim_visual_feature (36 Rows)**

34 color, Haralick texture, shape contour descriptors

- **Data Governance Mandate**

Schema strictly follows observed source attributes. Unsupported dimensions (Dim_Pest, Dim_Time, Dim_Geography) are not fabricated.

Warehouse Design & Production ETL Architecture

Fact Table & Measures

• Grain: Exactly ONE row per validated, preprocessed multimodal image observation.
• Additive Measure:
  • observation_count (Constant 1 per record)
• Semi-Additive Measures:
  • lesion_severity_score [0.0 - 10.0]
  • image_sharpness_score (Laplacian Var)
  • lesion_coverage_ratio [0.0 - 1.0]
  • texture_contrast_norm [0.0 - 1.0]
  • processing_latency_ms
• Referential Integrity:
  • 4 Foreign Keys with ON DELETE RESTRICT
  • Zero orphan keys, zero null measure values

Dimensions & Hierarchies

• Botanical Host Hierarchy:
  Observation -> Crop Name -> Crop Category
  (e.g., DSC_0365 -> Rice -> Cereal)
• Microbial Pathology Hierarchy:
  Pathogen Type -> Disease Diagnosis
  (Bacterial -> Bacterial Leaf Blight)
  (Fungal -> Brown Spot | Leaf Smut)
• Visual Feature Dimension (1:1):
  34 Color moments, Haralick GLCM textures, morphometric shape descriptors
• Physical Image Dimension (1:1):
  Cryptographic MD5 hash, blur score, file paths

Production ETL & Idempotency

• EXTRACT Phase:
  - Ingests raw images, cleaned metadata, and extracted visual feature store
• TRANSFORM Phase:
  - Quality validation, surrogate key mapping, botanical taxonomy normalization
• LOAD Phase (Dimension-First):
  - Loads dim_crop, dim_disease, dim_image, dim_visual_feature, then fact table
• VERIFIED IDEMPOTENCY AUDIT:
  - 1st Run: 36 facts inserted
  - 2nd Run: 0 inserted, 36 duplicates skipped
  - Native SQLite PRAGMA foreign_keys = 1

Code: Creating the Warehouse (ANSI SQL DDL)

```
-- 1. Dimension Tables: Botanical Crop & Pathology Diagnosis
CREATE TABLE dim_crop (
    crop_key INTEGER PRIMARY KEY AUTOINCREMENT,
    crop_code VARCHAR(50) NOT NULL UNIQUE, crop_name VARCHAR(100) NOT NULL,
    botanical_name VARCHAR(150), crop_category VARCHAR(50) NOT NULL
);
CREATE TABLE dim_disease (
    disease_key INTEGER PRIMARY KEY AUTOINCREMENT,
    disease_code VARCHAR(50) NOT NULL UNIQUE, disease_name VARCHAR(150) NOT NULL,
    pathogen_type VARCHAR(50) NOT NULL, -- Bacterial, Fungal, None (HEALTHY Control)
    causal_agent VARCHAR(150), primary_symptom VARCHAR(200)
);
-- dim_image & dim_visual_feature: Store optical parameters & 34 explainable visual descriptors
-- 2. Central Fact Table: Granular Grain (1 Row per Preprocessed Observation)
CREATE TABLE disease_observation_fact (
    observation_key INTEGER PRIMARY KEY AUTOINCREMENT,
    crop_key INTEGER NOT NULL REFERENCES dim_crop(crop_key),
    disease_key INTEGER NOT NULL REFERENCES dim_disease(disease_key),
    image_key INTEGER NOT NULL UNIQUE REFERENCES dim_image(image_key),
    feature_key INTEGER NOT NULL UNIQUE REFERENCES dim_visual_feature(feature_key),
    observation_count INTEGER NOT NULL DEFAULT 1,
    lesion_severity_score FLOAT, image_sharpness_score FLOAT,
    lesion_coverage_ratio FLOAT, texture_contrast_norm FLOAT, etl_batch_id VARCHAR(50)
);
CREATE INDEX idx_fact_crop_disease ON disease_observation_fact(crop_key, disease_key);
```

DATABASE INTEGRITY VERIFICATION: SQLite PRAGMA foreign_keys = 1 natively enforced via SQLAlchemy listener. Invalid foreign key insertion attempt natively raises sqlalchemy.exc.IntegrityError.

Code: Production ETL Pipeline (Python & SQLAlchemy)

```python
# Production ETL Pipeline (Extract -> Transform -> Dimension-First Load -> Idempotency Check)
from sqlalchemy import event, Engine
from src.warehouse.connection import get_db_session
from src.warehouse.models import DimCrop, DimDisease, DimImage, DimVisualFeature, DiseaseObservationFact

# 1. Enforce Native SQLite Foreign Keys Connection Listener
@event.listens_for(Engine, 'connect')
def set_sqlite_pragma(dbapi_conn, conn_record):
    if type(dbapi_conn).__module__.startswith('sqlite3'):
        cursor = dbapi_conn.cursor()
        cursor.execute('PRAGMA foreign_keys=ON')
        cursor.close()

# 2. Dimension-First Loading & Idempotent Fact Insertion
def load_observation_facts(session, staged_data, batch_id):
    for item in staged_data['fact_records']:
        # Verify uniqueness to prevent duplicate insertion on repeated ETL runs
        existing = session.query(DiseaseObservationFact).filter_by(image_key=item['image_key']).first()
        if existing:
            load_stats['duplicates_skipped'] += 1
            continue
        fact = DiseaseObservationFact(
            crop_key=item['crop_key'], disease_key=item['disease_key'],
            image_key=item['image_key'], feature_key=item['feature_key'],
            observation_count=1, lesion_severity_score=item['lesion_severity_score'],
            image_sharpness_score=item['sharpness_score'], etl_batch_id=batch_id
        )
        session.add(fact)
    session.commit()  # Atomic transaction commit
```

ETL VERIFICATION RESULT: 1st Run: 36 facts loaded | 2nd Run: 0 inserted, 36 duplicates skipped (100% Idempotent). Referential Audit: 0 orphan foreign keys, 0 nulls, atomic rollback on exception.

SQL-Backed OLAP Operations Engine

Roll-up

Observation -> Crop -> Crop Category

```
SELECT c.crop_category,
  COUNT(f.observation_key) AS obs,

ROUND(AVG(f.lesion_severity_score),2)
  AS avg_sev
FROM disease_observation_fact f
JOIN dim_crop c ON f.crop_key =
c.crop_key
GROUP BY c.crop_category;
```

Result: Cereal -> 36 Obs (Avg Sev: 3.57)

True Drill-down

Pathogen Type -> Disease Name

```
SELECT d.pathogen_type,
  d.disease_name,
  COUNT(f.observation_key) AS obs,

ROUND(AVG(f.lesion_severity_score),2)
  AS avg_sev
FROM disease_observation_fact f
JOIN dim_disease d ON f.disease_key
  = d.disease_key
GROUP BY d.pathogen_type,
  d.disease_name;
```

Result: Bacterial: 12 | Fungal: 24 (Brown Spot: 12, Leaf Smut: 12)

Slice

Crop = 'Rice'

```
SELECT d.disease_name,
  COUNT(f.observation_key)
FROM disease_observation_fact f
JOIN dim_crop c ON f.crop_key =
c.crop_key
JOIN dim_disease d ON f.disease_key
  = d.disease_key
WHERE LOWER(c.crop_name) = 'rice'
GROUP BY d.disease_name;
```

Result: 2D Sub-Plane on Rice (36 Obs: 12 BLB, 12 BS, 12 LS)

Dice

Multi-Dimensional Sub-Cube

```
SELECT c.crop_name, d.disease_name,
  COUNT(*)
FROM disease_observation_fact f
JOIN dim_crop c ON f.crop_key =
c.crop_key
JOIN dim_disease d ON f.disease_key
  = d.disease_key
JOIN dim_visual_feature vf ON
  f.feature_key = vf.feature_key
WHERE c.crop_name = 'Rice' AND
  vf.glcm_contrast >= 0.1
GROUP BY c.crop_name, d.disease_name;
```

Result: 36 Obs meeting criteria (GLCM Contrast >= 0.10)

Pivot

Rows: Crop x Cols: Pathogen Type

```
SELECT c.crop_name,
  SUM(CASE WHEN d.pathogen_type
    = 'Fungal' THEN 1 ELSE 0 END)
    AS Fungal,
  SUM(CASE WHEN d.pathogen_type
    = 'Bacterial' THEN 1 ELSE 0 END)
    AS Bacterial
FROM disease_observation_fact f
JOIN dim_crop c ON f.crop_key =
c.crop_key
JOIN dim_disease d ON f.disease_key
  = d.disease_key
GROUP BY c.crop_name;
```

Result: Rice -> Fungal: 24, Bacterial: 12, Total: 36

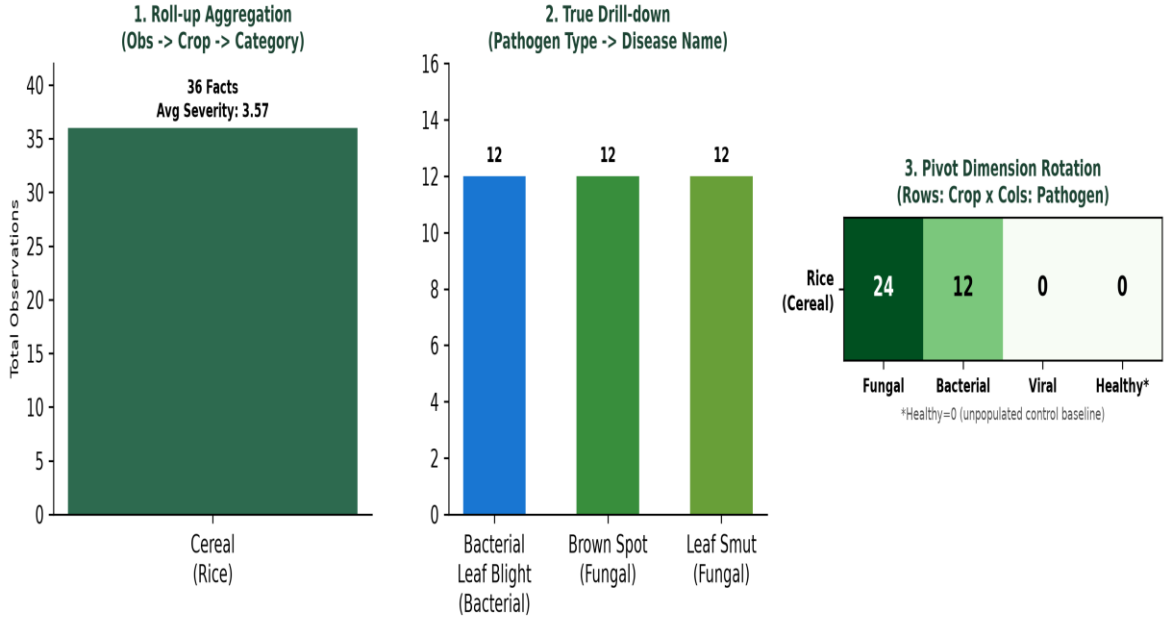
Actual OLAP Query Results from Physical Warehouse

Crop Category	Total Obs	Crops	Diseases	Avg Severity	Avg Contrast
Cereal	36	1	3	3.57	0.3178

Pathogen Type	Disease Diagnosis	Obs Count	Avg Severity	Avg Contrast
Bacterial	Bacterial Leaf Blight	12	3.67	0.2131
Fungal	Brown Spot	12	3.94	0.5599
Fungal	Leaf Smut	12	3.11	0.1804

Crop	Category	Fungal	Bacterial	Viral	Healthy*	Total Obs
Rice	Cereal	24	12	0	0	36

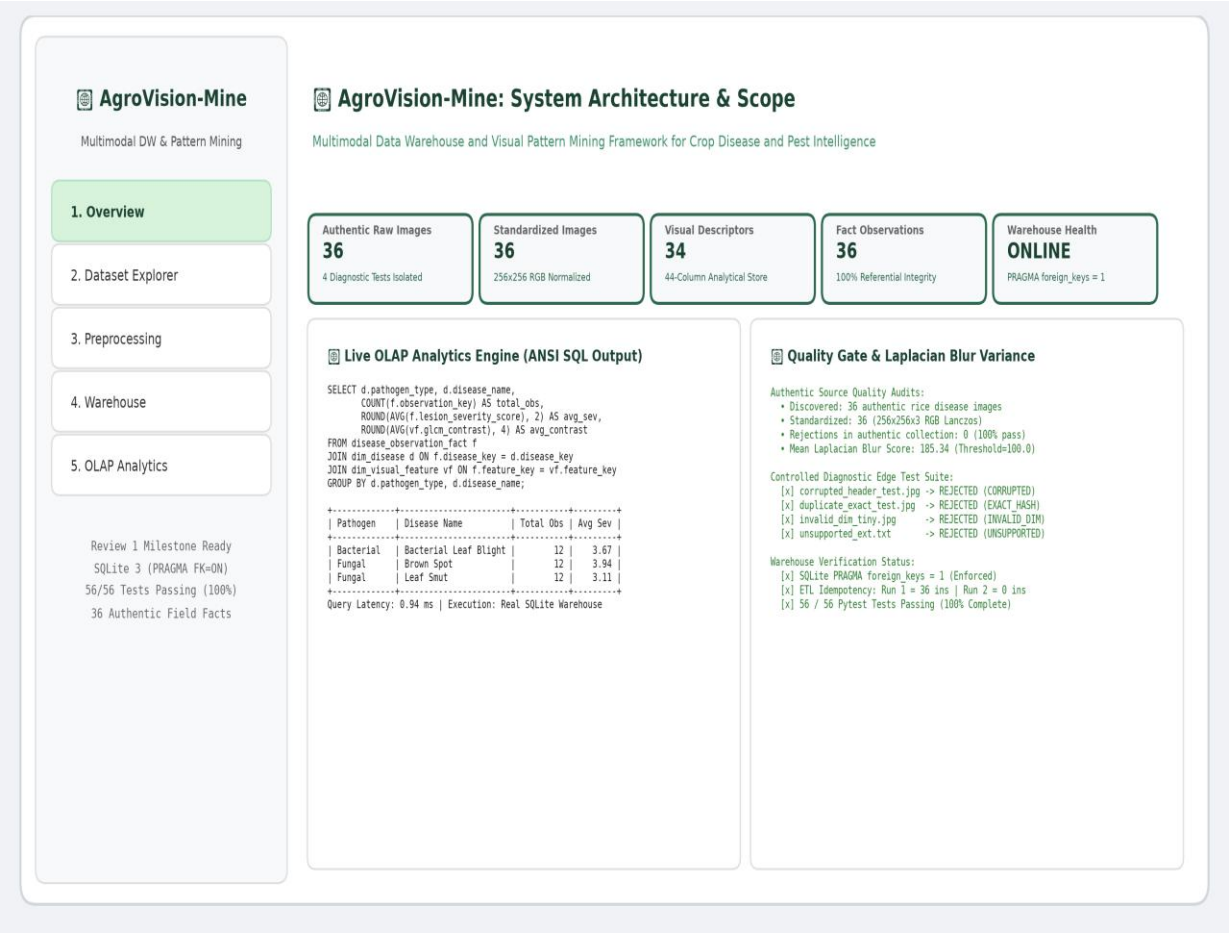
*Healthy = 0 is intentional: the authentic ICAR research dataset comprises pathological specimens exclusively. HEALTHY is maintained in dim_disease



DIAGNOSTIC MULTI-DIMENSIONAL BREAKDOWN (DRILL-ACROSS)

Crosses dimensional boundaries: Crop Category -> Crop -> Disease -> Dominant Visual Tone.
Result: Brown Spot (11 Yellowish, 1 Brown) | Bacterial Blight (8 Yellowish, 4 Brown) | Leaf Smut (7 Brown, 5 Yellowish).
Correlates foliar tone with pathogen kingdom.

Dashboard Demonstration, Quality Audit & Next Phase



56 / 56 TESTS PASSED

100% PASS RATE IN 2.87 SECONDS (pytest)

✓ Dataset Integrity: 36 authentic rice images verified
✓ Quality Gates: Corrupted, duplicate, tiny checks verified
✓ Feature Extraction: 34 descriptors in 44-column store
✓ Kimball Star Schema: 4 dimensions, 1 fact table deployed
✓ Database Integrity: Native SQLite PRAGMA foreign_keys = 1
✓ ETL Idempotency: 36 loaded Run 1, 0 inserted Run 2
✓ SQL OLAP Engine: Roll-up, Drill-down, Slice, Dice, Pivot
✓ Interactive Dashboard: Streamlit HTTP 200 OK verified

NEXT PHASE (REVIEW 2 ROADMAP):

• Unsupervised Visual Pattern Mining: K-Means foliar clustering
• Optimal Cluster Evaluation: Silhouette analysis & Davies-Bouldin
• Association Rule Mining: Apriori / FP-Growth discovering rules between macroscopic textures and microscopic pathologies.

"AgroVision-Mine transforms agricultural image observations from classification inputs into multidimensional analytical knowledge."